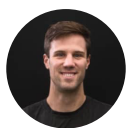


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Time to Peak Velocity

Time to peak velocity is a useful metric for highlighting how rapidly an athlete can accelerate a load. This reveals an athlete's rate of force development (RFD) and even their reactive strength index (RSI) helping coaches program effectively



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Time to peak velocity is a new rep metric added to the Metric app in version 5.9.

Time to peak velocity measures how rapidly an athlete can accelerate a load from its start position to the moment of peak velocity during the

concentric phase of a repetition. While peak velocity shows the ceiling of an athlete's speed capacity, time to peak reveals the rate at which they can access that speed—a critical component for explosive movements, particularly in Olympic lifting and dynamic effort training.

This metric quantifies an athlete's rate of force development (RFD) on any lift, providing coaches with direct feedback on explosive qualities that complement traditional velocity and power measurements. Time to peak is also a valuable metric to help make programming decisions and exercise selection in order to focus training on developing RFD.

The Science Behind Rate of Force Development

Rate of force development (RFD) represents the nervous system's ability to rapidly recruit motor units and generate force in the shortest time possible. This quality underpins explosive athletic movements—from the initial drive in Olympic lifts to the explosive power needed in jumping and throwing.

RFD is often more predictive of athletic performance than maximum strength alone. Elite athletes in court and field sports so often distinguish themselves not just by how much force they can ultimately produce, but by how quickly they can access and produce that force. In explosive movements.

The great news is that RFD is highly trainable, and thanks to barbell velocity tracking tools like the Metric app, RFD is easy to track over time through propulsive velocity and time to peak metrics — key data points to guide programming decisions.

Why Time to Peak Velocity Complements Existing Metrics

Metric offers a number of traditional and more modern ways to track and measure lifting performance, time to peak velocity is a newer metric available in the app that adds a more focused way of understanding and drilling even deeper into explosive power production in the weight room.

Mean Velocity

Historically the most popular and commonly used metric in barbell tracking, mean velocity provides a useful summary of overall lift speed, fatigue levels, and strength progression.

While reliable and easy to track, mean velocity is not without flaws. By averaging velocity across the entire concentric portion of the repetition mean velocity does not account for periods where the barbell may float or deceleration is applied mid-lift. An athlete might achieve high mean velocity through sustained grinding and cheating or flicking the barbell at the top of a repetition, rather than explosive early acceleration.

Mean velocity is a great general bar speed metric, however it provides less value for exercises aimed at developing RFD.

Peak Velocity

Peak velocity tells us the maximum speed achieved on a rep, but it doesn't tell us how long it took to reach this value. Two athletes might have identical peak velocities but one did it with a long steady build up, while the other snapped rapidly and achieved their peak velocity in

30% less time. This can leave coaches confused as the second athlete will clearly look more explosive, yet their peak values are the same.

Peak velocity is also a “noisy” data point and prone to measurement error, barbell vibrations, weight shifts, jerks and pulses may be enough to cause a brief peak velocity spike that doesn’t truly reflect explosive capacity.

Time to peak velocity helps with both these issues by providing context for when that peak occurred, rewarding intent, while also distinguishing between genuine power production and measurement artifacts.

Propulsive Velocity & Time to Peak

Mean propulsive velocity addresses many of these concerns by focusing only on the accelerating portion of the lift— which is why its the default starting velocity metric in Metric.

Using propulsive velocity/power and time to peak velocity in combination when training for power and RFD is a peanut butter and jelly combo of data points, offering reliable insights into how much power an athlete generates and how quickly they are able to unleash all of their force into accelerating the barbell.

Primary Applications and Benefits

Time to peak velocity can be used on any exercise, but it is best used on explosive movements with light to moderate loads, where rapid force development is paramount.

Olympic Lifting and Derivatives

Olympic lifts and their variations rely heavily on explosive initiation and rapid force development. Time to peak velocity helps assess and develop the explosive qualities critical for:

- Clean and snatch variations
- Power variations of Olympic lifts
- Hang positions and derivatives
- Push press and jerk movements

For these movements, shorter time to peak values indicate superior ability to rapidly accelerate loads from static or hanging positions.

Dynamic Effort Training

During power development phases (sometimes called speed-strength in legacy terminology) using lighter loads (30-60% 1RM), time to peak velocity provides direct feedback on explosive intent and development. This is particularly useful for:

- Dynamic effort squats and deadlifts
- Speed bench press and bench throw on a smith machine
- Explosive accessory movements

Exercise and Variation Specificity

Critical consideration: Time to peak velocity is highly specific to exercise variation, load lifted and starting position. Coaches should establish separate baselines for different variations (and probably for

each individual) rather than generalizing across movements.

Starting Position Impact

The starting position dramatically influences time to peak velocity values, especially on olympic lift variations:

Hang Clean Variations:

- Mid-thigh hang: 0.10–0.15 seconds (typical)
- Below-knee hang: 0.15–0.25 seconds (typical)
- Full clean from floor: 0.25–0.40 seconds (typical)

Each variation requires different time to reach peak velocity due to the distinct muscle lengths, joint angles, and mechanical advantages at each starting position.

Exercise-Specific Considerations

Excellent Applications:

- Pause box squats (static start)
- Pause bench press (eliminates bounce)
- Hang Olympic lift variations
- Power pull-ups from dead hang
- Push press and jerks

Limited Applications:

- Conventional deadlifts (can encourage jerky initiation and a lose

of tension/position)

- Touch-and-go bench press (stretch-shortening cycle interference and cheating)
- Continuous tempo squats (no static start)

Individual variation Tracking

Coaches should maintain separate time to peak velocity profiles for each of their athletes. While comparing between athletes can be an interesting way of contrasting your more elastic dominant athletes from more force-dominant athletes, time to peak is quite variable between athletes with range of motion playing a big role in your scores.

A more useful score would be to create a ratio between pause and countermovement variations of the same exercise to find an athletes individual elastic potential and RFD as a modified reactive strength index test.

More on this RSI assessment protocol later in the article.

Limitations and Considerations

Movement Quality Concerns

Time to peak velocity can inadvertently encourage jerky or violent movement initiation, particularly in exercises where smooth acceleration is preferred. Athletes may sacrifice technique quality in pursuit of faster time to peak values.

Risk Exercises:

- Conventional deadlifts (encourages yanking off the floor)
- Heavy squats (promotes explosive unracking rather than controlled descent)
- Technical Olympic lift learning phases (may disrupt timing)

Load Sensitivity

Time to peak velocity is highly sensitive to load changes. Small increases in weight can significantly impact values, making it important to compare results only at similar relative intensities.

Fatigue Sensitivity

This metric is one of the first to deteriorate under neuromuscular fatigue, making it useful for monitoring readiness but requiring careful interpretation when athletes are not fully recovered. A fatigued athlete will have slower bar speeds, longer time to peak velocity, and lower power production compared to their own recent training history when in a fatigued state.

How Time to Peak Velocity is Calculated

The calculation requires precise velocity tracking throughout the concentric phase to identify the exact moment of peak velocity achievement.

The Methodology

Step 1: Identify Concentric Phase StartThe system identifies when velocity transitions from zero (or negative) to positive, indicating the athlete has begun applying upward force to the barbell.

Step 2: Track Velocity Throughout Concentric PhaseContinuous velocity monitoring captures the acceleration curve as the athlete develops force and increases barbell speed.

Step 3: Identify Peak Velocity MomentThe system identifies when velocity reaches its maximum value for that repetition.

Step 4: Calculate Duration

Time to Peak Velocity = Peak_Velocity_Timestamp –
Concentric_Start_Timestamp

This provides the duration (in seconds) from movement initiation to peak velocity achievement.

Baseline Values and Interpretation

Values vary significantly based on exercise selection, load, starting position and athlete height/limb length. Establish exercise-specific and individual specific baselines rather than attempting to generalize across movements.

Typical Ranges by Exercise Category

Hang Olympic Lift Variations (60–80% 1RM)

- Elite Athletes: 0.08–0.18 seconds

- Intermediate Athletes: 0.12–0.25 seconds
- Novice Athletes: 0.18–0.35 seconds

Pause Squat Variations (50–70% 1RM)

- Elite Athletes: 0.15–0.30 seconds
- Intermediate Athletes: 0.25–0.45 seconds
- Novice Athletes: 0.35–0.60 seconds

Pause Bench Press (50–70% 1RM)

- Elite Athletes: 0.12–0.25 seconds
- Intermediate Athletes: 0.20–0.40 seconds
- Novice Athletes: 0.30–0.55 seconds

Practical Implementation

Exercise-Specific Baselines

Establish separate baselines for each major exercise variation using loads at or near the point of maximum power production (typically 40–75% 1RM). Test 3–5 repetitions across 3–5 weights and find baseline values. Over time you should be able to achieve faster time to peak velocity times on all weights.

Modified Reactive Strength Index Assessment

Time to peak velocity enables a practical assessment of an athlete's

elastic vs. strength qualities by comparing performance between countermovement and pause variations of the same exercise. This modified reactive strength index reveals whether athletes are elastic-dominant (are able to utilise the stretch-shortening cycle well) or strength-dominant (generate similar power from static positions and have poor elastic qualities).

Test Protocol

Exercise Selection: Choose exercises where both countermovement and pause variations are practical.

- Squat jump vs. pause squat (barbell or trapbar)
- touch and go bench vs. pause bench press (bench throws are usually better than standard bench if available)
- Hang clean off rack vs dip and drive hang clean
- Push press vs strict press

Testing Procedure:

1. **Warm-up thoroughly** with both exercise variations
2. **Select same load for both variations** (typically 40–60% 1RM where peak power occurs)
3. **Perform pause variation first:** 2–3 reps with 2–3 second pause, maximum explosive intent
4. **Rest 3–5 minutes** to ensure full recovery
5. **Perform countermovement variation:** 2–3 reps with natural countermovement, maximum explosive intent
6. **Record best (fastest) time to peak velocity** for each variation

Calculation and Interpretation

Reactive Strength Index Ratio:

RSI Ratio = Time to Peak (Pause) / Time to Peak
(Countermovement)

Interpretation:

- **RSI Ratio > 1.3:** Elastic-dominant athlete (significant benefit from stretch-shortening cycle)
- **RSI Ratio 1.1–1.3:** Balanced athlete (moderate elastic benefit)
- **RSI Ratio < 1.1:** Strength-dominant athlete (minimal elastic contribution)

Training Implications

Elastic-Dominant Athletes (High RSI Ratio):

- Emphasize plyometric and countermovement training
- Use stretch-shortening cycle exercises
- Monitor countermovement variations for progress tracking

Strength-Dominant Athletes (Low RSI Ratio):

- Focus on pause variations and static strength development
- Incorporate elastic training to develop this quality
- Use pause variations for primary assessment

Example Results:

- Athlete A: Pause squat 0.28s, Countermovement squat 0.18s → RSI Ratio: 1.56 (Elastic-dominant)
- Athlete B: Pause squat 0.22s, Countermovement squat 0.20s → RSI Ratio: 1.10 (Strength-dominant)

This assessment provides valuable insight into an athlete's neuromuscular profile and helps guide exercise selection and training emphasis.

Note on Programming With RSI Results

It's important to also consider an athlete's general strength levels before making programming adjustments from an RSI test, strength is the rising tide that improves all other qualities downstream of it, so regardless of RSI score a weak athlete (under 1x bodyweight squat or equivalent standard) will almost always benefit from getting stronger regardless of RSI score! This is especially relevant for athletes whose sporting training is already providing plenty of plyometric and elastic type training, they are more likely to benefit from using their gym time to fill their strength deficiency as a top priority before worrying about specialised power training.

Programming Applications for Time to Peak Velocity

Olympic Lifting Development Monitor time to peak velocity during hang variations to assess explosive development from static positions.

Target progressive reduction while maintaining peak velocity.

Dynamic Effort Phases Use time to peak velocity to ensure explosive intent during speed–strength training. Maintain consistent or improving values across training blocks.

Exercise Selection Guidance Choose exercises that naturally lend themselves to time to peak assessment—pause variations, hang positions, and static start movements provide the most reliable and actionable data.

Conclusion

Time to peak velocity provides valuable insight into rate of force development during explosive barbell exercises. This metric works well when combined with propulsive velocity measurements by quantifying the rate of force development (RFD), particularly valuable for Olympic lifting, explosive movements and dynamic effort or power training.

Time to peak velocity provides a useful metric to assess athletic elasticity via the reactive strength index (RSI) test. This RSI score can be a useful way to group athletes and highlight deficiencies and tendencies in their athletic profile.

The exercise–specific nature of time to peak velocity requires coaches to establish separate baselines for different variations and starting positions. When applied appropriately to suitable exercises, this metric offers direct feedback on explosive qualities that can guide training decisions and monitor neuromuscular adaptations.

Understanding both the applications and limitations of time to peak velocity ensures coaches can integrate this metric effectively into comprehensive training monitoring protocols.

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